LEARNER GUIDE

Basic Urban Farming with Hydroponics

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Document Version Control Record

Version	Date	Description	Approved by
1.0	24 March 2025	Legacy approved course package baseline.	Course Development Team
2.0	29 August 2026	Substantial technical rebuild aligned to current Singapore context, farm tour and hydroponics-kit activity.	Dr Alfred Ang

Table of Contents

Course and Competency Alignment

- TSC: AGR-FOP-2002-1.1 Urban Farming Implementation and Management Control.
- LO1: Perform farming activities and equipment operations according to standard operating procedures.
- LO2: Execute farming operations in accordance with regulations, workplace safety and health, food-safety and housekeeping requirements.
- Knowledge: equipment and machinery; farming SOPs; GAP/food safety/WSH; cleaning and maintenance.
- Abilities: prepare; operate to targets; comply; clean, maintain and store equipment.

How to Use This Guide

The slide deck explains mechanisms and decisions. This Learner Guide contains the detailed operating sequence. Always follow the site-specific SOP, product label, Safety Data Sheet and trainer instructions where they are more restrictive. Stop when a required control is absent.

1. Hydroponic Root-Zone Control

Hydroponics replaces soil functions

Water carries dissolved ions; air spaces or aeration supply oxygen; channels/media support roots; monitoring replaces much of soil buffering.

The operating variables

Record pH, EC, solution temperature, level/volume, flow and visible root condition. Interpret them together.

Mass balance

Water uptake and evaporation can raise EC; selective uptake can change ion proportions even when EC looks stable.

Plant demand

Light, heat, humidity, airflow, crop stage and canopy size change water and nutrient demand.

2. Systems and Equipment

NFT

A shallow recirculating film provides water and oxygen but has low tolerance to pump interruption.

DWC

Roots hang in aerated solution; air delivery and hygiene are critical.

Kratky

A static reservoir develops an air gap; inappropriate refilling can remove that oxygen zone.

Ebb-and-flow, drip and media

Irrigation duration, drainage and air-filled porosity must match the crop and media.

Pump and drainage

Verify delivered flow at operating head, unobstructed returns, freeboard and power-off behaviour.

3. Nutrient and Water Management

Source water

Baseline pH, EC and temperature; investigate alkalinity when pH continually rebounds.

Nutrient mixing

Keep Part A and Part B concentrates separate; add each to water with circulation.

pH and EC

Use crop-specific bands. EC is total ionic strength, not element-by-element analysis.

Calibration

Use fresh standard, rinse correctly, record calibration and verify the instrument response.

Correction

Add small measured amounts, circulate, remeasure, record before/action/after.

4. Crop, Environment and Hygiene

Nursery and transplant

Accept only uniform, healthy plugs with established roots and intact media.

Light and climate

Use PPFD/DLI, temperature, humidity/VPD and airflow trends to explain crop response.

IPM

Prevent, monitor, identify, intervene lawfully, and verify. Sticky cards are trend tools.

Root diagnosis

Inspect colour, odour, firmness, branching, flow, oxygen risk and chemistry before nutrient correction.

Cleaning and maintenance

Remove residue before disinfection; follow concentration/contact time; rinse/dry; function-test before return to service.

5. Singapore Cases and Current Context

Food Story 2

By 2035 Singapore targets local production capacity equivalent to 20% of fibre and 30% of protein needs. In 2025 output was about 8% and 25%, respectively.

ComCrop

Rooftop hydroponics integrates sunlight, supplemental light, monitoring and manual crop operations.

Sustenir

Controlled-environment vertical NFT demonstrates high area productivity and recirculation.

Singrow

Cultivar development, controlled environment and automation target temperate crops in tropical conditions.

Greenphyto

The 2026 multi-storey project demonstrates the scale-and-automation thesis; reported capacity must be separated from realised output.

6. Aquaponics

Coupled biology

Fish, nitrifying microbes and plants share water and risk.

Nitrogen conversion

Ammonia is converted to nitrite then nitrate; ammonia and nitrite are critical hazards.

Oxygen and pH

Fish, roots and microbes all need oxygen; pH must balance animal health, nitrification and nutrient availability.

Solids and biofilter

Remove solids before they foul roots; protect attached nitrifiers from disinfectants and abrupt changes.

System choice

Aquaponics adds biological products and biological complexity; decoupling can separate optimisation at higher equipment and control cost.

Detailed Procedure A — Farm Tour Evidence Walk

1. **Attend the farm briefing and confirm permitted areas, photography rules and emergency instructions.**

Why: The farm is a live workplace and may contain commercial information.

Acceptance evidence: Your checklist records the briefing and permission state.

2. **Trace one water path from reservoir to delivery, root zone and return without touching equipment.**

Why: Flow-path understanding reveals system boundaries and single points of failure.

Acceptance evidence: A labelled flow sketch with inlet, return, dosing/sample point and overflow.

3. **Compare crop condition at inlet, middle and return positions and across tiers where present.**

Why: Spatial patterns help distinguish shared-input from local flow/light issues.

Acceptance evidence: Three concise observation notes and permitted photographs where allowed.

4. **Record how pH, EC, temperature and level/flow are measured and corrected.**

Why: An operating control requires instrument, setpoint, frequency, action and verification.

Acceptance evidence: Completed water/nutrient control row with observation versus operator claim labelled.

5. **Inspect WSH, food safety, pest monitoring, housekeeping and maintenance evidence.**

Why: Safe compliant production depends on controls beyond plant growth.

Acceptance evidence: At least one existing control and one improvement opportunity, with no personal data.

6. **Review the checklist with a peer, resolve unclear statements and submit the PDF evidence set.**

Why: Peer review improves factual clarity and separates inference from observation.

Acceptance evidence: Signed checklist, flow sketch and only permitted photographs.

Detailed Procedure B — Commission and Operate a Hydroponics Kit

7. **Read the kit SOP; identify pump, reservoir, channel, tubing, return, planting sites and electrical isolation.**

Why: Correct component identification prevents unsafe assembly and misrouting.

Acceptance evidence: Component check complete; no damaged or unknown parts.

8. **Inspect for cleanliness, stability, cable damage, tubing kinks and safe separation of water and power.**

Why: Visual acceptance prevents leaks, collapse and electrical exposure.

Acceptance evidence: Pre-use inspection signed; defects quarantined and reported.

9. **Assemble only as shown in the SOP, fill with source water, keep electrical connectors dry and check for leaks before energising.**

Why: Dry inspection and controlled filling limit spill and electrical consequence.

Acceptance evidence: Stable leak-free kit before power-on.

10. **Energise the pump and confirm delivery to every planting site, unobstructed return and adequate reservoir freeboard.**

Why: The farthest site and return behaviour establish functional flow.

Acceptance evidence: Flow verification recorded; no overflow or abnormal noise.

11. **Measure and record source-water pH, EC and temperature using calibrated meters.**

Why: Baseline separates source-water chemistry from nutrient addition.

Acceptance evidence: Time-stamped values with units, meter IDs and calibration status.

12. **Add the trainer-specified Part A dose to water, circulate, then add Part B and circulate; never mix concentrates directly.**

Why: Separation prevents concentrated calcium/phosphate/sulfate precipitation.

Acceptance evidence: No visible precipitate; dose and mixing times recorded.

13. **Measure EC; make only small trainer-approved corrections with circulation between additions.**

Why: Incremental correction reduces overshoot.

Acceptance evidence: EC is within the activity setpoint; before/action/after log complete.

14. **Measure pH; if adjustment is authorised, use the labelled product, PPE and dedicated dosing tool, then circulate and remeasure.**

Why: Acid/alkali dosing is a chemical task with a stop condition.

Acceptance evidence: pH is within setpoint and all additions are recorded.

15. **Place an accepted seedling so its plug is supported, roots meet moisture and the crown remains stable.**

Why: Correct placement supports root access without stem/crown damage.

Acceptance evidence: Seedling label, centred plug and intact healthy root evidence.

16. **Conduct the trainer-controlled power-off/restart check only where spill and electrical controls are adequate.**

Why: Commissioning must verify drain-down, reservoir capacity and restart recovery.

Acceptance evidence: No overflow/siphon issue; flow returns to every site after restart.

17. **Record final values and obtain trainer verification before draining.**

Why: The finished state must be demonstrably controlled.

Acceptance evidence: Final pH/EC/temperature/flow record and trainer check.

18. **Isolate power, drain through the approved route, remove residue, clean, rinse, dry, reassemble and store.**

Why: Cleaning and maintenance prevent contamination and preserve equipment readiness.

Acceptance evidence: Dry complete kit, chemicals secured, maintenance/defect log completed.

Troubleshooting Decision Table

Observation	Check first	Do not assume	Escalation/acceptance
Wilting with full reservoir	Flow, root condition, solution temperature, aeration	That more water or fertiliser is automatically needed	Restore flow/oxygen safely; isolate disease risk; record recovery
EC rises as level falls	Water loss, top-up history, instrument verification	That the nutrient recipe is too strong	Follow SOP top-up/correction and verify after mixing
pH repeatedly rises	Source alkalinity, new media, uptake pattern	That one large acid dose is safe	Use incremental correction; escalate persistent rebound
Brown slimy roots	Temperature, DO/aeration, flow, odour, shared loop	That leaf feeding solves the root cause	Protect reservoir, isolate per SOP, sanitise only as approved
Uneven plants along channel	Delivered flow, slope, blockage, light uniformity	That seed genetics is the only cause	Correct physical non-uniformity and verify endpoint flow

References and Current Sources

https://www.tertiarycourses.com.sg/wsq-basic-urban-farming-with-hydroponics.html

https://www.sfa.gov.sg/news-publications/newsroom/singapore-food-statistics-2025

https://www.mse.gov.sg/latest-news/oral-reply-to-parliamentary-question-on-revised-local-production-targets/

https://www.sfa.gov.sg/food-science-and-technology/technology/agriculture-technologies

https://www.sfa.gov.sg/fromSGtoSG/farms/farm/Detail/comcrop

https://www.sfa.gov.sg/fromSGtoSG/farms/farm/Detail/sustenir-agriculture

https://www.sfa.gov.sg/food-for-thought/article/detail/hydroponics-getting-to-the-root-of-the-myths

https://www.sfa.gov.sg/recognition-programmes-grants/grants/agri-food-cluster-transformation-act-fund

https://enterprise.nus.edu.sg/startup-story/singrow-2/

https://www.businesstimes.com.sg/singapore/smes/singapore-food-output-gets-s80-million-boost-opening-greenphytos-vertical-farm-tallest-world https://www.rhs.org.uk/vegetables/hydroponics

https://extension.illinois.edu/blogs/good-growing/2026-01-16-growing-without-soil-getting-started-hydroponics-home

https://extension.unr.edu/publication.aspx?PubID=2756